CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 23: Graphical Representation (Histograms and Ogives)

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Find the class mark and the class size of the class 25–35.

2. Convert the classes 1–10, 11–20, 21–30 into continuous classes.

3. The class marks of a distribution are 12, 18, 24, 30. Find the class intervals.

4. Classes 0–5, 5–10, 10–15, 15–20 have frequencies 3, 7, 9, 5. Write the cumulative frequencies.

5. For the data in the previous question, write the points that are plotted to draw the ogive.

6. In a histogram, the tallest rectangle is on the class 30–40. What is the modal class, and between which values does the mode lie?

7. An ogive is drawn for 80 students. At which cumulative frequencies are the median, Q_1 and Q_3 read?

8. From an ogive, $Q_1 = 32$ and $Q_3 = 58$. Find the inter-quartile range.

9. An ogive for 200 workers shows that 150 workers earn less than ₹500 a day. How many earn ₹500 or more? What percentage is this?

10. In a test taken by 120 students, the top 25% are to be given a prize. At which cumulative frequency should the ogive be read to find the lowest mark that wins a prize?

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Answer Key

1. Class mark 30; class size 10
2. 0.5–10.5, 10.5–20.5, 20.5–30.5
3. 9–15, 15–21, 21–27, 27–33
4. 3, 10, 19, 24
5. (0, 0), (5, 3), (10, 10), (15, 19), (20, 24)
6. 30–40; between 30 and 40
7. 40, 20 and 60
8. 26
9. 50 workers; 25%
10. $120 - 30 = 90$

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